How-To Library

Informatica

Performance tuning guidelines for Microsoft Azure Cosmos DB SQL API Connector

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Abstract

You can tune the Informatica mapping parameters to optimize the performance of Microsoft Azure Cosmos DB SQL API Connector. This article describes general reference guidelines to help you tune the performance of Microsoft Azure Cosmos DB SQL API Connector.

Supported Versions

- Informatica Cloud® Data Integration Microsoft Azure Cosmos DB SQL API Connector

Table of Contents

Overview. 2
Performance tuning parameters. 2
Tune the reader performance. 2
Tune the writer performance. 3
Scalability test. 3
Reader scalibility test. 3
Writer scalibility test. 3

Overview

The performance tuning is an iterative process to analyze the performance, use guidelines to estimate and define parameters that impact the performance, and monitor and adjust the results as required.

This document describes the mapping parameters that you can tune to optimize the performance of Microsoft Azure Cosmos DB SQL API Connector.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of Microsoft Azure Cosmos DB SQL API Connector might vary based on individual environments and other parameters even when you use the same data.

Performance tuning parameters

You can optimize the performance of Microsoft Azure Cosmos DB SQL API mappings by tuning the reader and writer performance.

Tune the reader performance

Tune the reader performance to optimize the performance of Microsoft Azure Cosmos DB SQL API mappings.

Mappings in advanced mode

To improve the performance of a mapping that reads from a Microsoft Azure Cosmos DB SQL API source, switch the mapping to advanced mode.

When you read a TPC-H Lineitem Dataset - SF1, the performance improves by approximately 23 times in the advanced mode.

Tune the writer performance

Tune the writer performance to optimize the performance of Microsoft Azure Cosmos DB SQL API mappings.

Hierarchical data

- When you increase the hierarchy levels in the dataset from one to three, the performance decreases by 15%.
- When you increase the number of complex columns in the dataset from five to ten, the performance decreases by approximately 10%.

Mappings in advanced mode

To improve the performance of a mapping that writes to a Microsoft Azure Cosmos DB SQL API target, switch the mapping to advanced mode.

When you write a TPCH Lineitem Dataset - SF10, the performance improves by 87% in the advanced mode.

Partition key

The uniqueness of data in the partition key column impacts the writer performance.

The more unique the data in the partition key column the better the performance.

Scalability test

Use the scalability test to understand the effects of increasing the data volume and request units on the reader and writer performance.

Reader scalibility test

The reader scalability test describes the effect of increasing the data volume and request units on the execution time of a mapping when you read data from Microsoft Azure Cosmos DB SQL API.

- When you increase the data volume and keep the request units constant, the performance scales linearly.
- When you increase the data volume and request units at the same scale, the performance scales linearly.

Writer scalibility test

The writer scalability test describes the effect of increasing the data volume and request units on the execution time of a mapping when you write data to Microsoft Azure Cosmos DB SQL API.

- When you increase the data volume and keep the request units constant, the performance scales linearly.
- When you scale the request units and keep the data volume constant, the performance improves at the same scale.
- When you increase both the data volume in the source and the request units in Microsoft Azure Cosmos DB SQL API, the performance remains consistent.

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